ASSESSMENT - 3

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ASSESSMENT TOOL - 3

1. Intelligent Agents in a smart Home Automation System.

Introduction :

* An intelligent agent is a system that percives its environment through sensors and takes Action through Actuator to Achive some objectives.
* In a smart home, sensor detect motion, temperature, light intensity while Actuator control lights, air Conditioner and alarms.

1. Simple Reflex Agent :

* A simple reflex Agent selects Actions based only on the current percept using condition-Action Rules.

Advantages :

* Simple to design and implement.
* Responds quickly to real time inputs

Disadvantages :

* Does not remember past events.

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* Cannot learn over sequences.

2. Model Based Agent:

* A Model based Agent maintains an internal state of the environment using Current inputs and previous Observations.

Advantages:

* Handles partially Observable environments
* Uses past information to make better decisions

Disadvantages:

* Requires memory and a Model of the environment
* More complex than a simple reflex Agent.

3. Goal - Based Agent:

* A Goal Based Agent selects Actions that help Achieve a specific goal.

Eg: in smart home:

* Achieve a Comfortable room temperature of 24°c

Advantages:

* Achieve a comfortable room temperature of 24°c

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can choose different Action to Achieve the same goal

Disadvantages :

* Requires Searching or Planning to Select Action
* May Consume more Computational resources

4. **Utility - Based Agent :**

* A utility based Agent evaluates possible Action using a utility function that Measure how desirable each outcome

Advantages :

* Balances multiple Competing Objectives
* Supports trade-offs between Comfort, cost

Disadvantage :

* Designing a suitable utility function is difficult
* Requires more Computation and information
* Incorrect utility weights can produce undesirable decision.

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2. Uninformed Search Strategies for a Robot :

1. **Uniform cost search**

* Uniform cost search expands the node with the lowest total path cost, $g(n)$ from the starting position
* It uses priority queue to select the least-cost node

Working Procedure :

1. Place the starting position in the priority Queue with cost 0.
2. Remove the node with the lowest path cost
3. Check whether the node is exit
4. If not exit, explore unvisited Neighboring positions.
5. Calculate the cumulative cost
6. Continue until the exit found

Time and space complexity

Time : $O(b^{1 + \lfloor C^*/\varepsilon \rfloor})$

Space : $O(b^{1 + \lfloor C^*/\varepsilon \rfloor})$

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Uniformed Cost search:

* uninformed search Algorithm used in Artificial Intelligence to find the Least cost path from Starting node to Goal node
* It expands the node with the Lowest Cumulative path cost $g(n)$.
* It uses Priority Queue to select the Next node.
* It Considers the cost of every edge rather than the number of edges.
* It is similar to Dijkstra's Algorithm when searching for a goal in graph.

Important characteristics:

1. Search Strategy: Expands the Lowest-cost node first.
2. Evaluation function: $f(n) = g(n)$ where $g(n)$ is the total cost from the Start node to node n.
3. Data structure: Priority Queue (Minimum priority first)

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4. Completeness :

complete when step costs are bounded below by a positive constant.

5. optimality :

Finds the least cost solution when all step costs are Non-negative.

6. Hemistic.

Does not require a Hemistic function.

7. Application :

GPS Navigation, Network Routing, Robotics. transportation, planning.

8. Limitation.

It may explore many node and Consume significant time and Memory.

Advantages :

→ Finds the optimal least-cost path

→ Handles weighted graph

→ Does not need hemistic information

→ useful when action have different costs

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Iterative deepening search (IDS)

* Iterative Deepening search repeatedly perform depth limited depth first search, increasing the depth limit at each iteration.

Working procedure:

1. Begin with a depth limit of 0.
2. Search the maze upto the current depth limit
3. If the increase not found, increase the depth limit by 1
4. Repeat the search from the starting position
5. Stop when reach.

Eg: IDS searches depth 0, then depth 1, then depth 2.

Advantages:

* uses much less memory than VCS.
* complete when the branching factor is finite
* Finds a shallowest solution when each step has equal cost
* useful when the solution depth is unknown

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Disadvantages:

* Repeats the exploration of upper-level Nodes
* May take longer if the exit is very deep.
* Does not guarantee a Minimum-cost path, when Step cost differ.

Time and Space Complexity:

for Branching factor b and Shallowest solution depth d:

Time: $O(b^d)$

Space: $O(bd)$

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